IN 3240 Software Testing

Test management

Chapter 5 – Part 1

1. Test organization
2. Test planning and estimation

1. Test organization
- 1.1 Test organization and independence
- 1.2 Tasks of the test leader and tester

2. Test planning and estimation
- 2.1 Test planning
- 2.2 Test planning activities
- 2.3 Entry criteria
- 2.4 Exit criteria
- 2.5 Test estimation
- 2.6 Test strategy, test approach

3. Test progress monitoring and control
- 3.1 Test progress monitoring
- 3.2 Test reporting
- 3.3 Test control

4. Configuration management

5. Risk and testing
- 5.1 Project risks
- 5.2 Product risks

6. Incident management

Test organization and independence

Testing software and developing (building) software are not the same

- Different tasks involved
- Require different mindsets from testers and developers

Testing is an assessment of quality

- Assessments are not always positive

Separate the testers from the developers

- Improve defect finding by using independent testers
- Avoid author bias → Objective assessments

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Test organization and independence

- The effectiveness of finding defects by testing and reviews can be improved by using independent testers.
- Options for independence are:

1. No independent testers. Developers test their own code.
2. Independent testers within the development teams.
3. Independent test team or group within the organization, reporting to project management or executive management.
4. Independent testers from the business organization or user community.
5. Independent test specialists for specific test targets such as usability testers, security testers or certification testers (who certify a software product against standards and regulations).

** Independent testers outsourced or external to the organization. (Highest level of independence, but not so used in practice)*

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- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

Test organization and independence

- For large, complex or safety critical projects, it is usually best to have multiple levels of testing, with some or all of the levels done by independent testers.
- Development staff can participate in testing, especially at the lower levels

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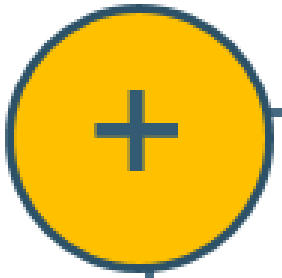
Advantages & disadvantages



Independent testers see other and different defects, and are unbiased.

An independent tester can verify assumptions people made during specification and implementation of the system.

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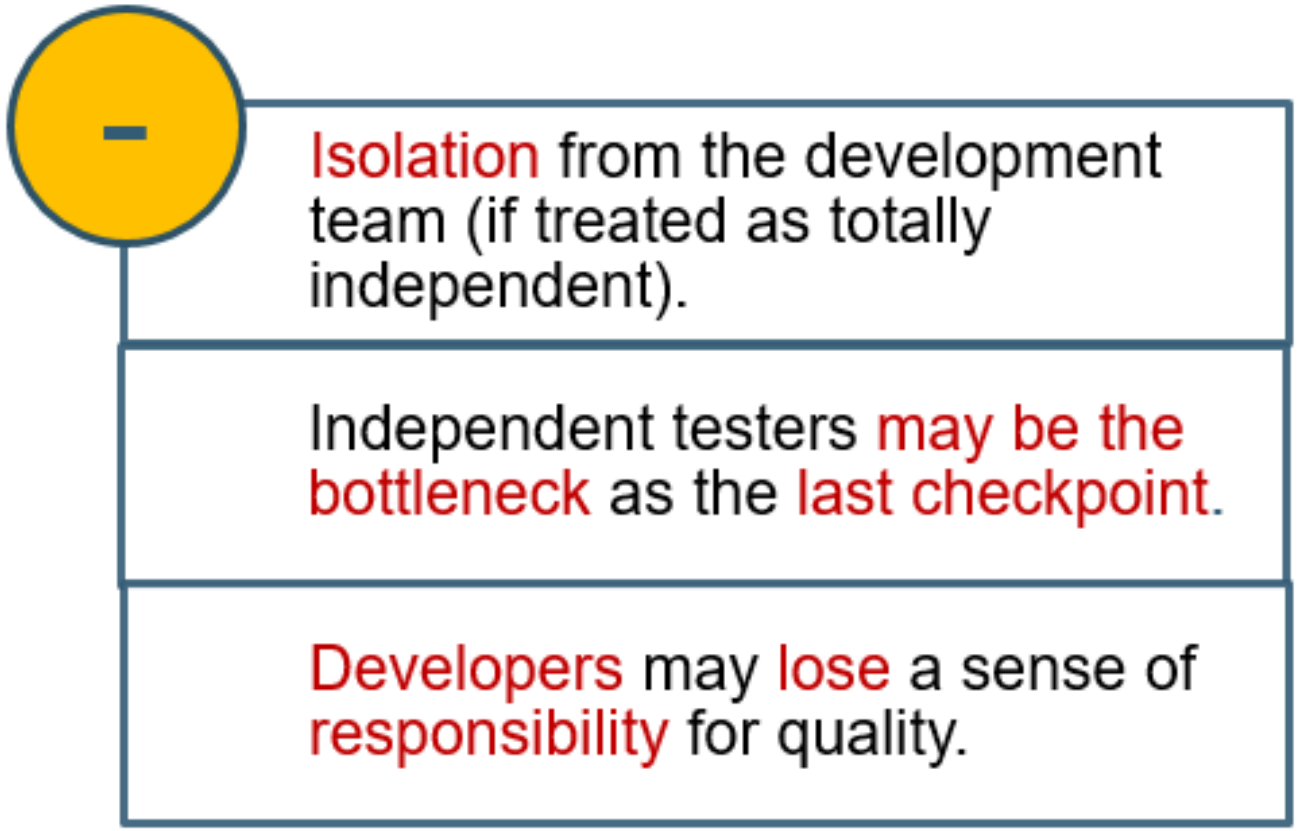

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Advantages & disadvantages

Note

Testing tasks may be done by people in a specific testing role, or may be done by someone in another role, such as:

- project manager
- quality manager
- developer
- business and domain expert
- infrastructure or IT operations

(this can be both good and bad)

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Tasks of the test leader and tester

There is wide variation in the roles that people within the test team play. The two most common roles are

- **test leader**
- **tester**

Test leader = test manager / test coordinator.
The test leader plans, monitors and controls the *testing activities and tasks*.

The tester
The tester reviews and contributes to test plan, analyses, designs, prepares, implements and executes tests.

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Tasks of the test leader

Coordination	of the test strategy and plan with project managers

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Tasks of the test leader

Coordination	of the test strategy and plan with project managers
Plan the tests	Understanding the test objectives and risks – including: • selecting test approaches • estimating the time, effort and cost of testing • acquiring resources • defining test levels, cycles • planning incident management

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Automation of tests	Decide what should be automated, to what degree, and how.
Select test tools	Select tools to support testing and organize trainings for tool users.
Test environment	Decide about the implementation of the test environment.
Test summary reports	Write test summary reports based on the information gathered during testing.

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Test automation	Automate tests (may be supported by a developer or a test automation expert).

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Test automation	Automate tests (may be supported by a developer or a test automation expert).
Other metrics	Measure performance of components and systems (if applicable).

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Test tools	Use test tools (for administration, management or monitoring) as required.
Test automation	Automate tests (may be supported by a developer or a test automation expert).
Other metrics	Measure performance of components and systems (if applicable).
Help the others	Review tests developed by others.

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- *1.1 Test organization and independence*
- *1.2 Tasks of the test leader and tester*

2. Test planning and estimation
- *2.1 Test planning*
- *2.2 Test planning activities*
- *2.3 Entry criteria*
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3. Test progress monitoring and control
- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

Skills

People involved in testing need

- **Application or business domain**: A tester must understand the intended behavior and the problem the system will solve in order to spot improper.
- **Technology**: A tester must be aware of issues, limitations and capabilities of the chosen implementation technology, in order to locate problems and the 'likely to fail' functions and features.
- **Testing:** A tester must know the testing topics discussed in this book - in order to carry out the test tasks assigned.

1. Test organization
- *1.1 Test organization and independence*
- *1.2 Tasks of the test leader and tester*

2. Test planning and estimation
- *2.1 Test planning*
- *2.2 Test planning activities*
- *2.3 Entry criteria*
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- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

Skills

People involved in testing

need basic professional and social qualifications such as

- literacy
- the ability to prepare and deliver written and verbal reports
- the ability to communicate effectively
- contribute to a positive relationship between developer and tester

1. Test organization
- *1.1 Test organization and independence*
- *1.2 Tasks of the test leader and tester*

2. Test planning and estimation
- *2.1 Test planning*
- *2.2 Test planning activities*
- *2.3 Entry criteria*
- *2.4 Exit criteria*
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3. Test progress monitoring and control
- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

Test planning and estimation

Who will do
what,
when
and how

1. Test organization
- *1.1 Test organization and independence*
- *1.2 Tasks of the test leader and tester*

2. Test planning and estimation
- *2.1 Test planning*
- *2.2 Test planning activities*
- *2.3 Entry criteria*
- *2.4 Exit criteria*
- *2.5 Test estimation*
- *2.6 Test strategy, test approach*

3. Test progress monitoring and control
- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

The purpose of a test plan

A test plan is the project plan for the testing work to be done.

- Writing a test plan forces us to confront the challenges that await us and focus our thinking on important topics.
- The test planning process and the plan itself serve as vehicles for communicating with other members of the project team, testers, peers, managers and other stakeholders.
- The test plan also helps us manage change. As we gather more information, we revise our plans.

1. Test organization
- *1.1 Test organization and independence*
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- *2.1 Test planning*
- *2.2 Test planning activities*
- *2.3 Entry criteria*
- *2.4 Exit criteria*
- *2.5 Test estimation*
- *2.6 Test strategy, test approach*

3. Test progress monitoring and control
- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

Test planning

Planning may be documented in:

- a project or master test plan
- and in separate test plans for test levels, such as integration testing, system testing and acceptance testing.

Test planning is a continuous activity and is performed in all life cycle processes and activities.

Feedback from test activities is used to recognize changing risks so that planning can be adjusted.

1. Test organization
- *1.1 Test organization and independence*
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- *2.1 Test planning*
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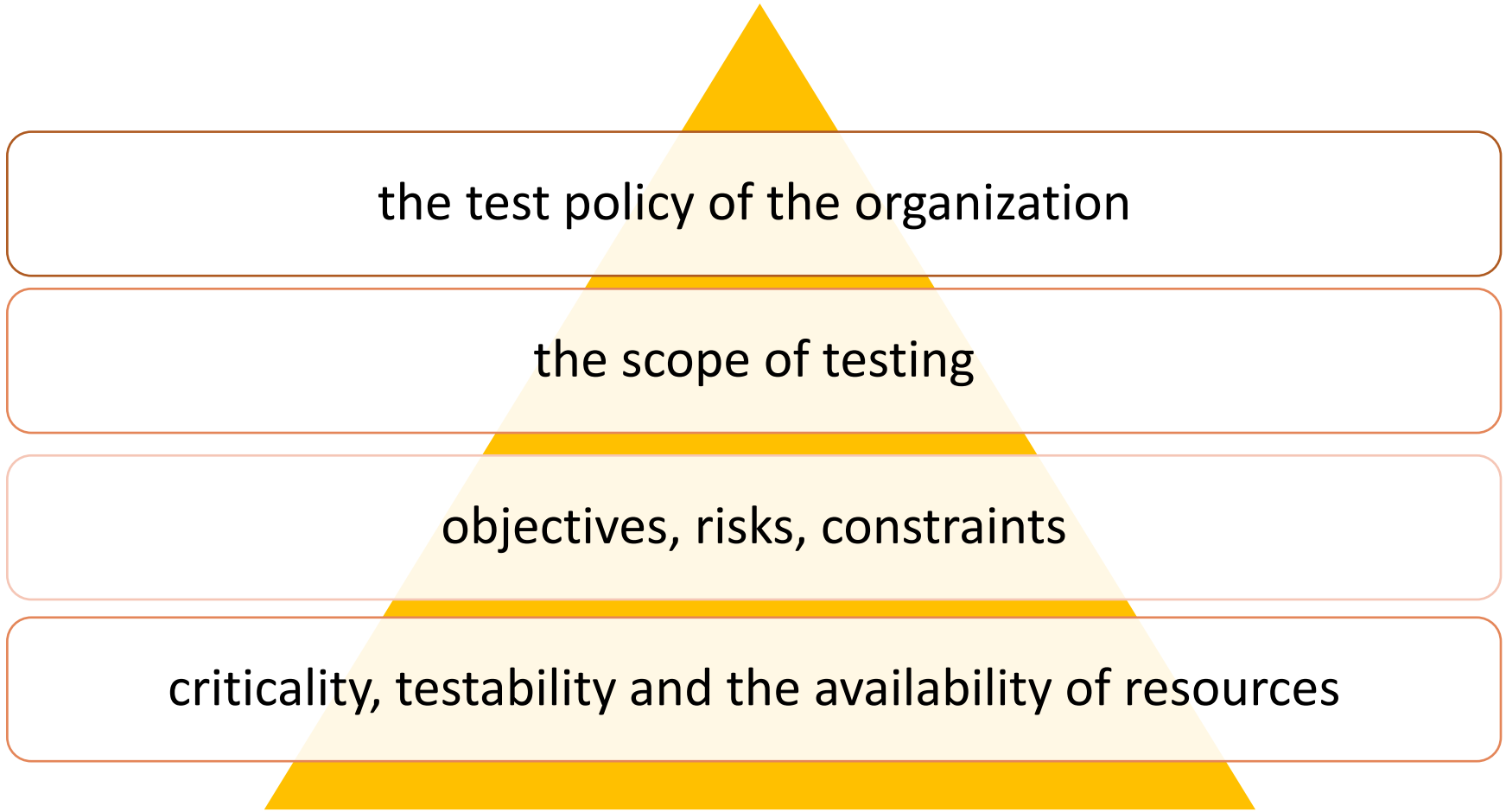
3. Test progress monitoring and control
- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

Test planning



** Outlines of test planning documents are covered by the 'Standard for Software Test Documentation' (IEEE 829).*

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- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

Test planning activities

Scope and risk	Determining the scope and risks of testing

1. Test organization
- *1.1 Test organization and independence*
- *1.2 Tasks of the test leader and tester*

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- *2.1 Test planning*
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- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

Test planning activities

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Objectives	Identifying the objectives of testing.

1. Test organization
- *1.1 Test organization and independence*
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- *2.1 Test planning*
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- *3.1 Test progress monitoring*
- *3.2 Test reporting*
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5. Risk and testing
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6. Incident management

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Overall approach	Defining the overall approach of testing, including: • the definition of the test levels • entry and exit criteria.

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- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

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- *5.1 Project risks*
- *5.2 Product risks*

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- *3.1 Test progress monitoring*
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5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

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- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

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- *1.2 Tasks of the test leader and tester*

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- *2.1 Test planning*
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- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

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Resources	Assigning resources for the different activities defined.

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- *1.1 Test organization and independence*
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- *2.1 Test planning*
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- *2.4 Exit criteria*
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- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

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Schedule	Scheduling test analysis and design activities. Scheduling test implementation, execution and evaluation
Resources	Assigning resources for the different activities defined.
Metrics	Selecting metrics for monitoring and controlling test preparation and execution, defect resolution and risk issues.

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- *1.1 Test organization and independence*
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2. Test planning and estimation
- *2.1 Test planning*
- *2.2 Test planning activities*
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- *2.4 Exit criteria*
- *2.5 Test estimation*
- *2.6 Test strategy, test approach*

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- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

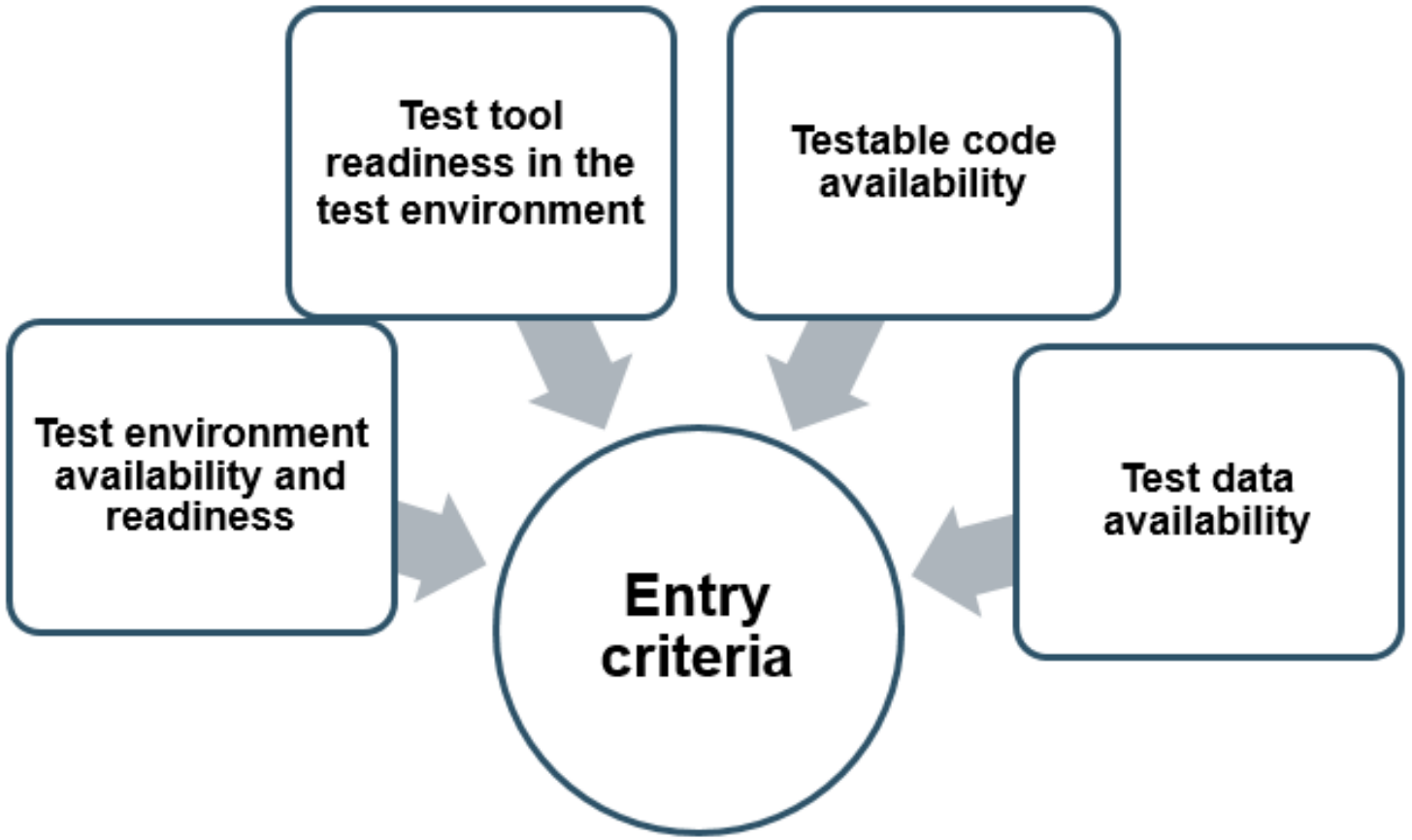
4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

Entry criteria

Entry criteria defines when to start testing.



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- *1.1 Test organization and independence*
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- *2.1 Test planning*
- *2.2 Test planning activities*
- *2.3 Entry criteria*
- *2.4 Exit criteria*
- *2.5 Test estimation*
- *2.6 Test strategy, test approach*

3. Test progress monitoring and control
- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

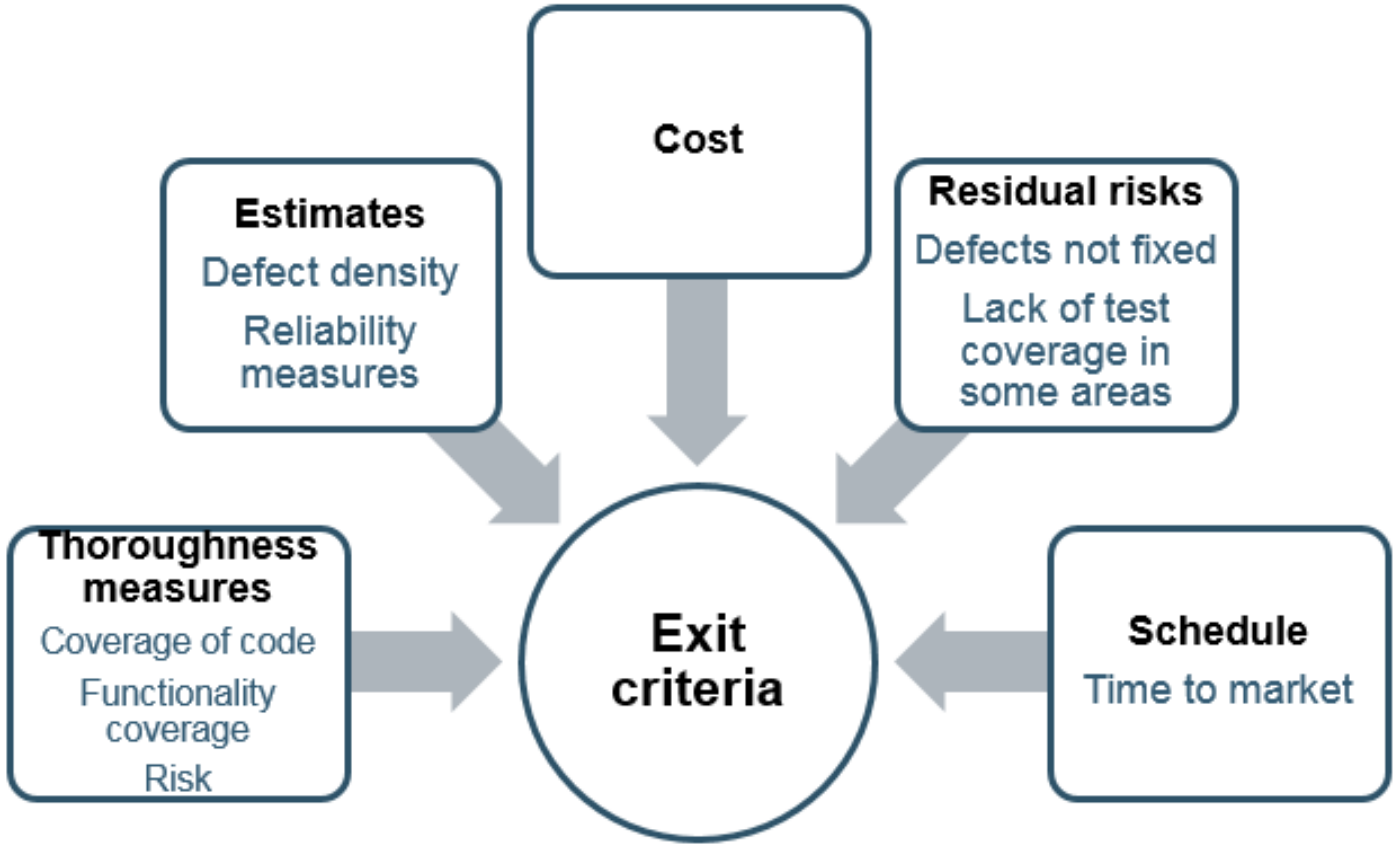
4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

Exit criteria

Exit criteria is to define when to stop testing, such as at the end of a test level, end of project or when a set of tests has a specific goal.



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- *2.1 Test planning*
- *2.2 Test planning activities*
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- *2.4 Exit criteria*
- *2.5 Test estimation*
- *2.6 Test strategy, test approach*

3. Test progress monitoring and control
- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

Test estimation

The testing work is usually a subproject within the larger project.

Fundamental techniques of estimation can be adapted for testing

We can break down a testing project into phases

- planning and control
- analysis and design
- implementation and execution
- evaluating exit criteria and reporting
- test closure

Within each phase we identify activities and within each activity we identify tasks and perhaps subtasks.

1. Test organization
- *1.1 Test organization and independence*
- *1.2 Tasks of the test leader and tester*

2. Test planning and estimation
- *2.1 Test planning*
- *2.2 Test planning activities*
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- *2.4 Exit criteria*
- *2.5 Test estimation*
- *2.6 Test strategy, test approach*

3. Test progress monitoring and control
- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

Test estimation

To identify the activities and tasks, we work both

- forward
- backward

To ensure accuracy of the estimate, make sure that you subdivide the work into tasks that are short in duration, say one to three days.

If they are much longer - say two weeks – there will be a risk that long and complex sub-tasks are 'hidden' within the larger task.

1. Test organization
- 1.1 Test organization and independence
- 1.2 Tasks of the test leader and tester

2. Test planning and estimation
- 2.1 Test planning
- 2.2 Test planning activities
- 2.3 Entry criteria
- 2.4 Exit criteria
- 2.5 Test estimation
- 2.6 Test strategy, test approach

3. Test progress monitoring and control
- 3.1 Test progress monitoring
- 3.2 Test reporting
- 3.3 Test control

4. Configuration management

5. Risk and testing
- 5.1 Project risks
- 5.2 Product risks

6. Incident management

Test estimation techniques

Two approaches for the estimation of test effort are covered in this syllabus

The metrics-based approach

Estimating the testing effort based on metrics of former or similar projects or based on typical values.

The expert-based approach

Estimating the tasks by the owner of these tasks or by experts.

1. Test organization
- *1.1 Test organization and independence*
- *1.2 Tasks of the test leader and tester*

2. Test planning and estimation
- *2.1 Test planning*
- *2.2 Test planning activities*
- *2.3 Entry criteria*
- *2.4 Exit criteria*
- *2.5 Test estimation*
- *2.6 Test strategy, test approach*

3. Test progress monitoring and control
- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

Test estimation

A good solution is to combine the two strategies:

- **First** create the work-breakdown structure and a detailed bottom-up estimate.

- **Second**; we then apply models and rules of thumb to check and adjust the estimate bottom-up and top-down using past history.

This approach tends to create an estimate that is both more accurate and more defensible than either technique by itself.

1. Test organization
- *1.1 Test organization and independence*
- *1.2 Tasks of the test leader and tester*

2. Test planning and estimation
- *2.1 Test planning*
- *2.2 Test planning activities*
- *2.3 Entry criteria*
- *2.4 Exit criteria*
- *2.5 Test estimation*
- *2.6 Test strategy, test approach*

3. Test progress monitoring and control
- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

Test estimation – an example

Performance tests need to be run in a special test environment that look like the production or field environment, due to the response time and resource utilization.

Performance testing involves

- special tools to generate load and check response
- environment acquisition and configuration
- time to identify and hire a performance test professional, if required
- training activities, if required

A detailed test plan for performance testing should be written in the test planning phase, where time is required to draft, review and finalize a performance test plan.

1. Test organization
- *1.1 Test organization and independence*
- *1.2 Tasks of the test leader and tester*

2. Test planning and estimation
- *2.1 Test planning*
- *2.2 Test planning activities*
- *2.3 Entry criteria*
- *2.4 Exit criteria*
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- *2.6 Test strategy, test approach*

3. Test progress monitoring and control
- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

Factors affecting the test effort

The testing effort may depend on a number of factors, including:

Product factors	• the quality of the specification (the test basis) • the size of the product • the complexity of the problem domain • the importance of non-functional quality e.g. usability, performance, security etc.
Process factors	• the development model • availability of test tools (e.g. test executing tools) • skills of the people involved • time pressure
The outcome of testing	• the number of defects • the amount of rework required

1. Test organization
- *1.1 Test organization and independence*
- *1.2 Tasks of the test leader and tester*

2. Test planning and estimation
- *2.1 Test planning*
- *2.2 Test planning activities*
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- *2.4 Exit criteria*
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- *2.6 Test strategy, test approach*

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- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

Test approaches and strategies

The test approach is the implementation of the test strategy for a specific project.

Since the test approach is specific to a project, the approach should be documented in the test plan

One way to classify test approaches or strategies is based on the point in time at which the bulk of the test design work is begun:

Preventative approaches

Tests are designed as early as possible

Reactive approaches

Test design comes after the software or system has been produced

1. Test organization
- *1.1 Test organization and independence*
- *1.2 Tasks of the test leader and tester*

2. Test planning and estimation
- *2.1 Test planning*
- *2.2 Test planning activities*
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- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

5. Risk and testing
- *5.1 Project risks*
- *5.2 Product risks*

6. Incident management

Test approaches and strategies

Analytical approaches	e.g. risk-based testing - testing is directed to areas of greatest risk, requirement-based testing

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- *1.1 Test organization and independence*
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- *2.1 Test planning*
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- *3.1 Test progress monitoring*
- *3.2 Test reporting*
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Model-based approaches	testing using statistical information about failure rates (such as reliability growth models)
Methodical approaches	e.g. failure-based (including error guessing and fault-attacks), experienced-based, check-list based, and quality characteristic based.

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Process- or standard-compliant approaches	e.g. specified by industry-specific standards or the various agile methodologies.

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Process- or standard-compliant approaches	e.g. specified by industry-specific standards or the various agile methodologies.
Dynamic and heuristic approaches	e.g. exploratory testing, execution & evaluation are concurrent tasks.

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- *3.3 Test control*

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6. Incident management

Test approaches and strategies

Analytical approaches	e.g. risk-based testing - testing is directed to areas of greatest risk, requirement-based testing
Model-based approaches	e.g. testing using statistical information about failure rates (such as reliability growth models)
Methodical approaches	e.g. failure-based (including error guessing and fault-attacks), experienced-based, check-list based, and quality characteristic based.
Process- or standard-compliant approaches	e.g. specified by industry-specific standards or the various agile methodologies.
Dynamic and heuristic approaches	e.g. exploratory testing, execution & evaluation are concurrent tasks.
Consultative approaches	e.g. test coverage is evaluated by domain experts outside the test team.

1. Test organization
- *1.1 Test organization and independence*
- *1.2 Tasks of the test leader and tester*

2. Test planning and estimation
- *2.1 Test planning*
- *2.2 Test planning activities*
- *2.3 Entry criteria*
- *2.4 Exit criteria*
- *2.5 Test estimation*
- *2.6 Test strategy, test approach*

3. Test progress monitoring and control
- *3.1 Test progress monitoring*
- *3.2 Test reporting*
- *3.3 Test control*

4. Configuration management

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Dynamic and heuristic approaches	e.g. exploratory testing, execution & evaluation are concurrent tasks.
Consultative approaches	e.g. test coverage is evaluated by domain experts outside the test team.
Regression-averse approaches	e.g. include reuse of existing test material, extensive automation of functional regression tests.

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Test approaches and strategies

The choice of test approaches and strategies is one powerful factor in the success of the effort and the accuracy of the test plans and estimates.

When we choose test strategies there are many factors to consider:

- Risks
- Skills
- Objectives
- Regulations
- Product
- Business